

Economics

Definition 1: Economics is a social science that focuses on the production, distribution, and consumption of goods and services, and analyzes the choices that individuals, businesses, governments, and nations make to allocate resources.

Definition 2: Economics is the study of scarcity and how it affects the use of resources, the production of goods and services, the growth of production and well-being over time, and many other important and complicated issues that affect society.

Key point:

- Economics is the study of how people allocate scarce resources for production, distribution, and consumption, both individually and collectively.
- The two branches of economics are microeconomics and macroeconomics.
- Economics focuses on efficiency in production and exchange.
- Gross Domestic Product (GDP) and the Consumer Price Index (CPI) are widely used economic indicators.

Meaning & Definition:

In the words of E. F. Brigham and J. L. Pappas Managerial Economics is “the applications of economics theory and methodology to business administration practice”.

M. H. Spencer and Louis Siegelman explain the “Managerial Economics is the integration of economic theory with business practice for the purpose of facilitating decision making and forward planning by management”.

It is clear, therefore, that managerial economics deals with economic aspects of managerial decisions as well as those managerial decisions, which have an economics content. Managerial economics may therefore, be defined as a body of knowledge, techniques and practices which give substance to those economic concepts which are useful in deciding the business strategy of a unit of management.

Managerial economics is designed to provide a rigorous treatment of those aspects of economic theory and analysis that are most useful for managerial decision analysis says J. L. Pappas and E. F. Brigham.

Managerial Economics, therefore, focuses on those tools and techniques, which are useful in decision-making.

Assuming humans have unlimited wants within a world of limited means, economists analyze how resources are allocated for production, distribution, and consumption.

Economics is the study of how things are made, moved around, and used. It looks at how people, businesses, governments, and countries choose to use their resources. Economics is the study of how people act, based on the idea that people act rationally and try to get the most value or benefit. Economics is the study of how work and business are run. Since there are many ways to use human labor and many ways to get resources, it is the job of economics to figure out which ways produce the best results.

In general, economics can be broken into two parts: macroeconomics, which looks at how the economy works, and microeconomics, which looks at how people and businesses work.

The study of **microeconomics** focuses on the choices of individuals and businesses, and **macroeconomics** concentrates on the behavior of the economy as a whole, on an aggregate level.

Microeconomics

Microeconomics studies how individual consumers and firms make decisions to allocate resources. Whether a single person, a household, or a business, economists may analyze how these entities respond to changes in price and why they demand what they do at particular price levels.

Microeconomics analyzes how and why goods are valued differently, how individuals make financial decisions, and how they trade, coordinate, and cooperate.

Within the dynamics of supply and demand, the costs of producing goods and services, and how labor is divided and allocated, microeconomics studies how businesses are organized and how individuals approach uncertainty and risk in their decision-making.

A person, a family, a business, a group, or the government can all make decisions independently. Microeconomics looks at different parts of human behavior to figure out how people react to changes in prices and why they want certain things at certain

prices. Microeconomics tries to explain why and how different things have different values, how people make financial decisions, and how they can trade, work together, and cooperate in the best way. Microeconomics looks at how supply and demand change over time and how well things are made, and how much they cost. It also looks at how people divide and share work, set up and run businesses, and deal with uncertainty, risk, and strategic game theory.

Macroeconomics

Macroeconomics is the branch of economics that studies the behavior and performance of an economy as a whole. Its primary focus is the recurrent economic cycles and broad economic growth and development.

It focuses on foreign trade, government fiscal and monetary policy, unemployment rates, the level of inflation, interest rates, the growth of total production output, and business cycles that result in expansions, booms, recessions, and depressions.

Using aggregate indicators, economists use macroeconomic models to help formulate economic policies and strategies.

There is a connection between microeconomics and macroeconomics. The sum of all microeconomic events makes up an aggregate macroeconomic event. But these two areas of economics use very different theories, models, and research methods that can make them seem to go against each other. Many economists study how to put together the basics of microeconomics with macroeconomics into theory and research.

KEY DIFFERENCES BETWEEN MICRO AND MACRO ECONOMICS

The points given below explain the difference between micro and macroeconomics in detail:

1. Microeconomics studies the particular market segment of the economy, whereas Macroeconomics studies the whole economy, that covers several market segments.
2. Micro economics stresses on individual economic units. As against this, the focus of macroeconomics is on aggregate economic variables.
3. While microeconomics is applied to operational or internal issues, environmental and external issues are the concern of macroeconomics.

4. Microeconomics deals with an individual product, firm, household, industry, wages, prices, etc., while Macroeconomics deals with aggregates like national income, national output, price level, etc.
5. Microeconomics covers issues like how the price of a particular commodity will affect its quantity demanded and quantity supplied and vice versa while Macroeconomics covers major issues of an economy like unemployment, monetary/ fiscal policies, poverty, international trade, etc.
6. Microeconomics determine the price of a particular commodity along with the prices of complementary and the substitute goods, whereas the Macroeconomics is helpful in maintaining the general price level.
7. While analyzing any economy, micro economics takes a bottom-up approach, whereas the macroeconomics takes a top-down approach into consideration

Basis For Comparison	Microeconomics	Macroeconomics
Meaning	The branch of economics that studies the behavior of an individual consumer, firm, family is known as Microeconomics.	The branch of economics that studies the behavior of the whole economy, (both national and international) is known as Macroeconomics.
Deals with	Individual economic variables	Aggregate economic variables
Business Application	Applied to operational or internal issues	Environment and external issues
Scope	Covers various issues like demand, supply, product pricing, factor pricing, production, consumption, economic welfare, etc.	Covers various issues like, national income, general price level, distribution, employment, money etc.
Importance	Helpful in determining the prices of a product along with the prices of factors of production (land, labor, capital, entrepreneur etc.) within the economy.	Maintains stability in the general price level and resolves the major problems of the economy like inflation, deflation, unemployment and poverty as a whole.
Limitations	It is based on unrealistic assumptions, i.e. In microeconomics it is assumed that there is a full employment in the society which is not at all possible.	It has been analyzed that 'Fallacy of Composition' involves, which sometimes doesn't proves true because it is possible that what is true for aggregate may not be true for individuals too.

What Is the Role of an Economist?

An economist studies the relationship between a society's resources and its production or output, and their opinions help shape economic policies related to interest rates, tax laws, employment programs, international trade agreements, and corporate strategies.

Economists analyze economic indicators, such as gross domestic product and the consumer price index to identify potential trends or make economic forecasts.

According to the Bureau of Labor Statistics, 36% of all economists in the United States work for a federal or state agency. Economists are also employed as professors, by corporations, or as part of economic think tanks.²

What Are Economic Indicators?

Economic indicators detail a country's economic performance. Published periodically by governmental agencies or private organizations, economic indicators often have a considerable effect on stocks, employment, and international markets, and often predict future economic conditions that will move markets and guide investment decisions.

Gross domestic product (GDP)

The gross domestic product (GDP) is considered the broadest measure of a country's economic performance. It calculates the total market value of all finished goods and services produced in a country in a given year. The Bureau of Economic Analysis (BEA) also issues a regular report during the latter part of each month.³ Many investors, analysts, and traders focus on the advance GDP report and the preliminary report, both issued before the final GDP figures because the GDP is considered a lagging indicator, meaning it can confirm a trend but can't predict a trend.

Retail sales

Reported by the Department of Commerce (DOC) during the middle of each month, the retail sales report is very closely watched and measures the total receipts, or dollar value, of all merchandise sold in stores.⁵ Sampling retailers across the country acts as a proxy of consumer spending levels. Consumer spending represents more than two-thirds of GDP, proving useful to gauge the economy's general direction.

Industrial production

The industrial production report, released monthly by the Federal Reserve, reports changes in the production of factories, mines, and utilities in the U.S. One measure included in this report is the capacity utilization ratio, which estimates the portion of productive capacity that is being used rather than standing idle in the economy.⁶ Capacity utilization in the range of 82% to 85% is considered "tight" and can increase the likelihood of price increases or supply shortages in the near term. Levels below 80% are interpreted as showing "slack" in the economy, which may increase the likelihood of a recession.

Employment Data

The Bureau of Labor Statistics (BLS) releases employment data in a report called the nonfarm payrolls on the first Friday of each month.⁷ Sharp increases in employment indicate prosperous economic growth and potential contractions may be imminent if significant decreases occur. These are generalizations and it is important to consider the current position of the economy.

Consumer Price Index (CPI)

The Consumer Price Index (CPI), also issued by the BLS, measures the level of retail price changes, and the costs that consumers pay, and is the benchmark for measuring inflation. Using a basket that is representative of the goods and services in the economy, the CPI compares the price changes month after month and year after year.⁸ This report is an important economic indicator and its release can increase volatility in equity, fixed income, and forex markets. Greater-than-expected price increases are considered a sign of inflation, which will likely cause the underlying currency to depreciate.

Economic problem

In economics, the main problem is that there are only so many resources and so many wants. Economics has also shown that people can't get everything they want. As time goes on, the more our needs are met, the more we want.

The fundamental problem in economics is the issue with the scarcity of resources but unlimited wants. Economics has also pointed out that a man's needs cannot be fulfilled. The more our needs are fulfilled, the more wants we develop with time. By definition, scarcity implies a limited quantity of resources.

The three basic problem of economics are:

- **What to produce:** This is the first question that needs to be answered if society is to allocate its resources efficiently. Of course, no society can sustain itself if all the resources are spent on defense, and none are spent on food production. This first and foremost question helps to identify a set of things that society needs to sustain itself in balance.
- **How to produce:** How should the factors of production be allocated to produce the required things? What would be the efficient way to make food, and what would be the efficient way to make cars? How much labor force is there in a community? How would these choices impact the affordability of the final product? All these questions are densely combined in one question - how to produce?
- **For whom to produce:** Last but not least, the question of who will be the final user of the things made is important. The choices made when answering the first of the three questions mean that the scarce resources were used to create a set of particular products. This implies that there may not be enough of one specific thing for everyone. Imagine a lot of resources were allocated for food production. This means that not everyone in that society can have a car.

Resource constraints

A resource constraint is any limitation or risk related to resources allocated to projects. Identifying these resource management restrictions is part of the project planning process. Resource constraints can disrupt your project and impede effective delivery.

Welfare maximization

The welfare maximization problem is an optimization problem studied in economics and computer science. Its goal is to partition a set of items among agents with different utility functions, such that the welfare – defined as the sum of the agents' utilities – is as high as possible.

1. Welfare maximization refers to the policy which looks after the welfare of the society and its people.
2. Its main focus is to provide opportunities to all people equitably.

3. This ensures that there is fair distribution of goods and services among the rich and the poor.

Scope of Economics

The scope of economics refers to its area of study. Managerial economics refers to its area of study. Managerial economics, provides management with a strategic planning tool that can be used to get a clear perspective of the way the business world works and what can be done to maintain profitability in an ever-changing environment. Managerial economics is primarily concerned with the application of economic principles and theories to five types of resource decisions made by all types of business organizations.

- a. The selection of product or service to be produced.
- b. The choice of production methods and resource combinations.
- c. The determination of the best price and quantity combination
- d. Promotional strategy and activities.
- e. The selection of the location from which to produce and sell goods or service to consumer.

The production department, marketing and sales department and the finance department usually handle these five types of decisions. The scope of managerial economics covers two areas of decision making.

a. Operational or Internal issues b. Environmental or External issues

a. Operational issues:

1. Demand Analyses and Forecasting: A firm can survive only if it is able to the demand for its product at the right time, within the right quantity. Understanding the basic concepts of demand is essential for demand forecasting. Demand analysis should be a basic activity of the firm because many of the other activities of the firms depend upon the outcome of the demand forecast.

Demand analysis provides:

1. The basis for analyzing market influences on the firms; products and thus helps in the adaptation to those influences.

2. Demand analysis also highlights for factors, which influence the demand for a product. This helps to manipulate demand. Thus demand analysis studies not only the price elasticity but also income elasticity, cross elasticity as well as the influence of advertising expenditure with the advent of computers, demand forecasting has become an increasingly important function of managerial economics.

2. Pricing and competitive strategy: Pricing decisions have been always within the preview of managerial economics. Pricing policies are merely a subset of broader class of managerial economic problems. Price theory helps to explain how prices are determined under different types of market conditions. Competitions analysis includes the anticipation of the response of competitions the firm's pricing, advertising and marketing strategies. Product line pricing and price forecasting occupy an important place here.

3. Production and cost analysis: Production analysis is in physical terms. While the cost analysis is in monetary terms cost concepts and classifications, cost-output relationships, economies and diseconomies of scale and production functions are some of the points constituting cost and production analysis.

4. Resource Allocation: Managerial Economics is the traditional economic theory that is concerned with the problem of optimum allocation of scarce resources. Marginal analysis is applied to the problem of determining the level of output, which maximizes profit. In this respect linear programming techniques has been used to solve optimization problems. In fact, linear programming is one of the most practical and powerful managerial decision making tools currently available.

5. Profit analysis: Profit making is the major goal of firms. There are several constraints here an account of competition from other products, changing input prices and changing business environment hence in spite of careful planning, there is always certain risk involved. Managerial economics deals with techniques of averting or minimizing risks. Profit theory guides in the measurement and management of profit, in calculating the pure return on capital, besides future profit planning.

6. Capital or investment analyses: Capital is the foundation of business. Lack of capital may result in small size of operations. Availability of capital from various sources like equity capital, institutional finance etc. may help to undertake large-scale operations. Hence efficient allocation and management of capital is one of the most important tasks of the managers. The major issues related to capital analysis are:

1. The choice of investment project
2. Evaluation of the efficiency of capital
3. Most efficient allocation of capital Knowledge of capital theory can help very much in taking investment decisions. This involves, capital budgeting, feasibility studies, analysis of cost of capital etc.

7. Strategic planning: Strategic planning provides management with a framework on which long-term decisions can be made which has an impact on the behavior of the firm. The firm sets certain long-term goals and objectives and selects the strategies to achieve the same. Strategic planning is now a new addition to the scope of managerial economics with the emergence of multinational corporations. The perspective of strategic planning is global. It is in contrast to project planning which focuses on a specific project or activity. In fact, the integration of managerial economics and strategic planning has given rise to be new area of study called corporate economics.

B. Environmental or External Issues: An environmental issue in managerial economics refers to the general business environment in which the firm operates. They refer to general economic, social and political atmosphere within which the firm operates. A study of economic environment should include:

- a. The type of economic system in the country.
- b. The general trends in production, employment, income, prices, saving and investment.
- c. Trends in the working of financial institutions like banks, financial corporations, insurance companies
- d. Magnitude and trends in foreign trade;
- e. Trends in labor and capital markets;
- f. Government's economic policies viz. industrial policy monetary policy, fiscal policy, price policy etc.

The social environment refers to social structure as well as social organization like trade unions, consumer's co-operative etc. The Political environment refers to the nature of state activity, chiefly states' attitude towards private business, political stability etc. The environmental issues highlight the social objective of a firm i.e.; the firm owes a responsibility to the society. Private gains of the firm alone cannot

be the goal. The environmental or external issues relate managerial economics to macroeconomic theory while operational issues relate the scope to micro economic theory. The scope of managerial economics is ever widening with the dynamic role of big firms in a society.

Production Possibilities Curve in Economics:

A production possibilities curve in economics measures the maximum output of two goods using a fixed amount of input. The input is any combination of the four factors of production: natural resources (including land), labor, capital goods, and entrepreneurship.

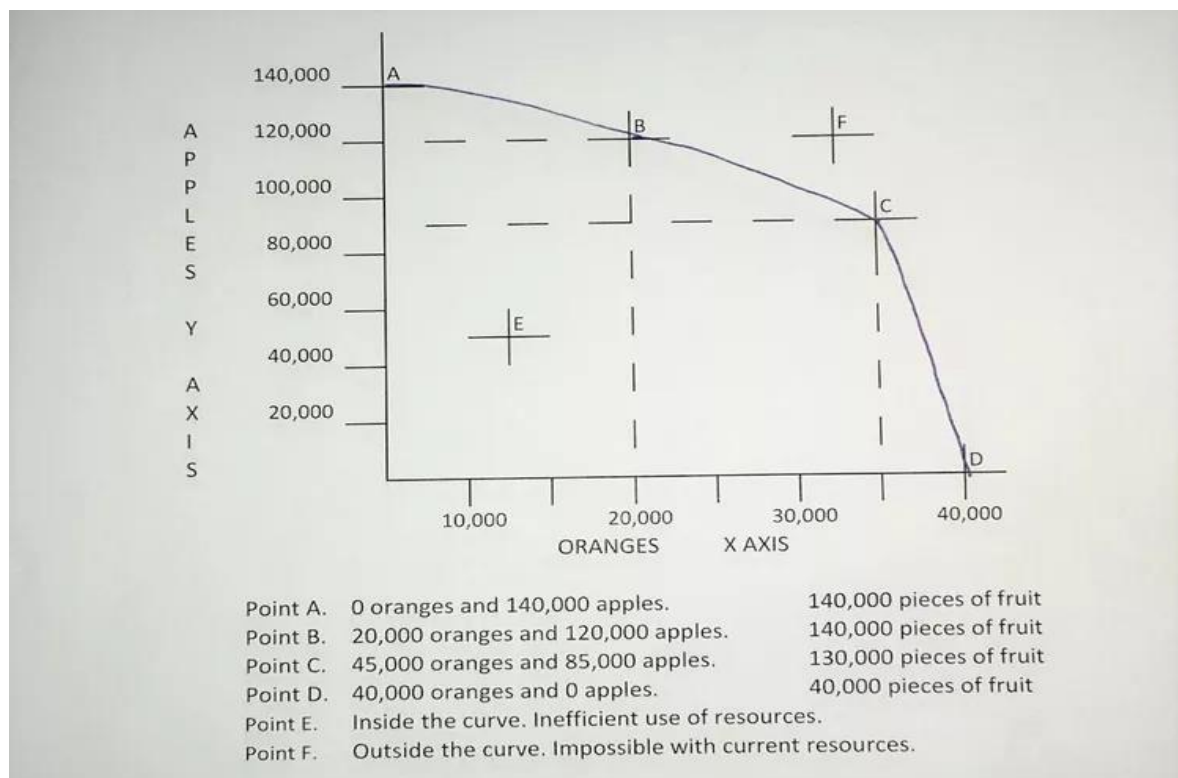
Definition and Examples of the Production Possibilities Curve:

In economics, the production possibilities curve is a visualization that demonstrates the most efficient production of a pair of goods. Each point on the curve shows how much of each good will be produced when resources shift to making more of one good and less of another.

The production possibilities curve measures the trade-off between producing one good versus another.

For example, say an economy produces 20,000 oranges and 120,000 apples. On the chart, that's point B. If it wants to produce more oranges, it must produce fewer apples. On the chart, Point C shows that if it produces 45,000 oranges, it can only produce 85,000 apples.

By describing this trade-off, the curve demonstrates the concept of opportunity cost. Making more of one good will cost society the opportunity of making more of the other good.



How the Production Possibilities Curve Works: The production possibility curve portrays the cost of society's choice between two different goods. An economy that operates at the production possibility frontier, or the very edge of this curve, has the highest standard of living it can achieve, as it is producing as much as it can be using its resources. If the amount produced is inside the curve, then all of the resources are not being used. On the chart above, that is point E.

One possible reason for such an inefficiency could be a recession or depression. If that occurs, there is not enough demand for either good. Layoffs may occur as well, resulting in lower levels of labor being used and therefore lowered production.

Other reasons for an inefficient production can be a bit more complicated. An economy will fall within the curve when it ignores its comparative advantage. For example, Florida has the ideal environment to grow oranges, and Oregon's climate is best for apples. Florida has a comparative advantage in orange production, and Oregon has one in apple production. If Florida ignored its advantage in oranges and tried to grow apples, it would create an inefficient use of resources. The U.S. economy would be operating within the curve, leading to a decrease in standard of living.

At the same time, any point outside the production possibilities curve is impossible. More of both goods cannot be produced with the limited resources. On the chart above, that is point F.

The production possibility curve bows outward. The highest point on the curve is when you only produce one good, on the y-axis, and zero of the other, on the x-axis. On the chart, that is Point A, where the economy produces 140,000 apples and zero oranges.

The widest point is when you produce none of the good on the y-axis, producing as much as possible of the good on the x-axis. On the chart, that is point D: The society produces zero apples and 40,000 oranges.

All the points in between are a trade-off of some combination of the two goods. An economy operates more efficiently by producing that mix. The reason is that every resource is better suited to producing one good over another. Some land is better suited for apples, while other land is best for oranges. Society does best when it directs the production of each resource toward its specialty. The more specialized the resources, the more bowed-out the production possibility curve.

How the Production Possibilities Curve Affects the Economy:?

The curve does not tell decision-makers how much of each good the economy should produce; it only tells them how much of each good they must give up if they are to produce more of the other good.² It is up to them to decide where the sweet spot is.

In a market economy, the law of demand determines how much of each good to produce. In a command economy, planners decide the most efficient point on the curve. They are likely to consider how best to use labor so there is full employment.

The leaders must create more demand for either or both products. Only after that occurs can more resources be used to produce greater output.

Supply-side economists believe the curve can be shifted to the right by simply adding more resources. However, without demand, they will only succeed in creating underutilized resources. There can be a benefit in increasing the labor force, though. Once the unemployed are working, they will increase demand and shift the curve to the right. For it to work, they must be paid enough to create the demand that shifts the curve outward. There must also be enough unemployed to make a

difference. An economy in full employment can't add more workers, no matter how much corporate taxes are cut.

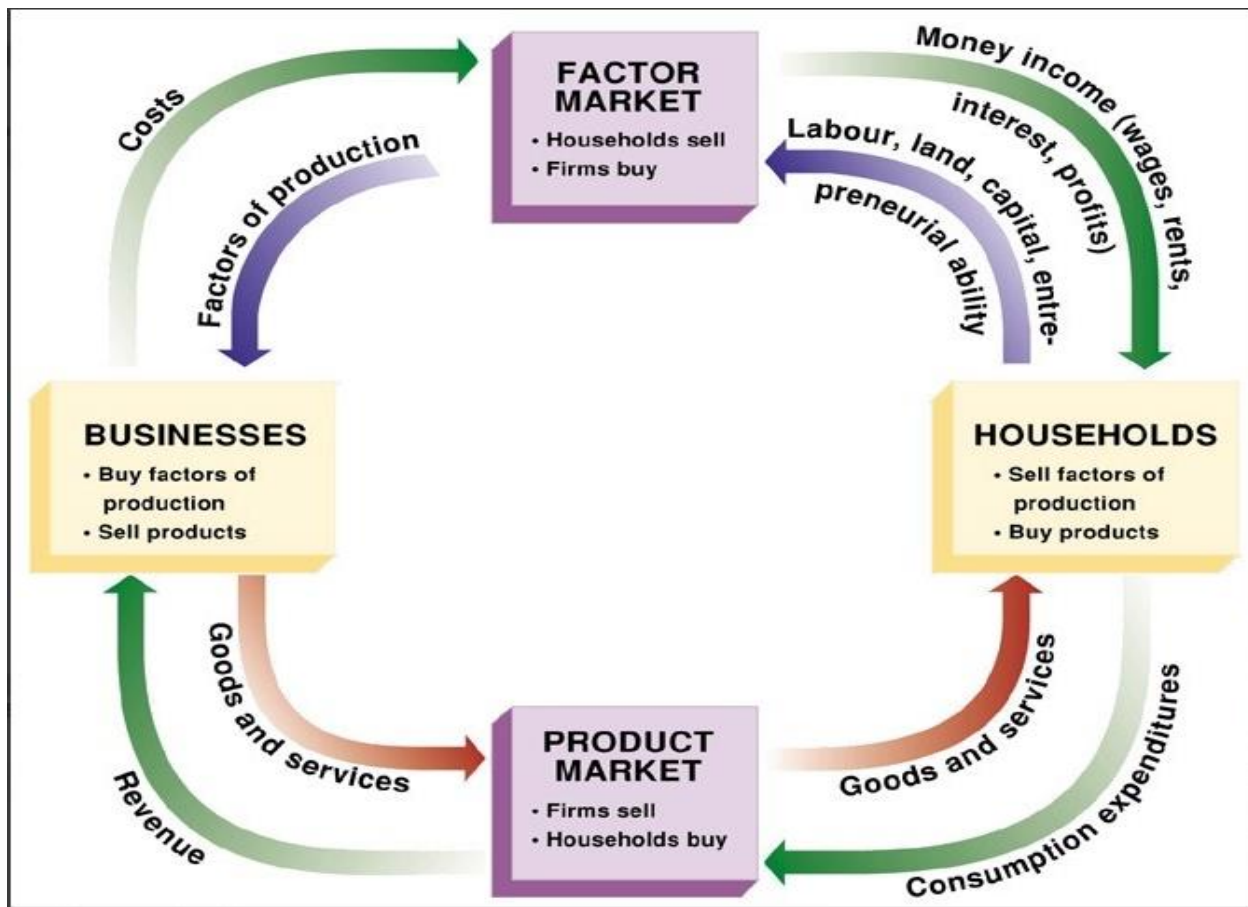
A decrease in resources can limit growth. If there is a shortage of one input, then more goods will not be produced, no matter how high the demand. In those situations, prices rise until demand falls to meet supply. It creates cost-push inflation.

Key Point

- The production possibilities curve shows the possible combinations of production volume for two goods using fixed resources.
- The assumption is that production of one commodity decreases if that of the other one increases.
- Production points inside the curve show that an economy is not producing at its comparative advantage, and production outside the curve is not possible.
- The production possibilities curve displays the right proportional mix of goods to be produced.

Circular flow of economic activities:

All market economies are characterized by a circular flow of economic activity. This means that money and products (including the products businesses need to operate) move in a circular fashion between businesses and households. This situation is often illustrated using a diagram that allows us to visualize the basic workings of the overall economy.



A market is any place or system allowing buyers and sellers to come together. A market economy is one in which the free interaction of buyers and sellers determines most of the important features of economic life. Most economic decisions in a market economy are based on the forces of supply (the amount of any good or service that a seller is willing to sell at a given price), demand (the amount of any good or service that buyers are willing to buy at a given price), and prices. Sellers tend to supply more and more of their products as prices rise (because they want to maximize their profits), while buyers tend to buy less and less of a product as prices rise (because they want to maximize their own economic well-being). When buyers decide to purchase or not purchase certain goods in certain quantities, sellers make corresponding decisions about what to supply and in what quantities.

The circular flow model illustrates how market forces determine the overall shape of the economy. Businesses and households act as both buyers and sellers in the economy. Businesses sell products to households in exchange for money, and households sell products called the factors of production (land, labor, and capital [money and equipment], the resources required to do business) to businesses. The

inner circle of the model shows products and services moving clockwise between businesses and households. The outer circle of the model shows income (money) moving counterclockwise between businesses and households.

This model is a simplification of economic activity, but it allows us to understand some central facts about market economies. Businesses and households are dependent on one another. They are bound to one another by supply and demand relationships that, in being played out, dictate the working of the economy. The money that households use to satisfy their needs and wants comes from businesses, and the money that businesses need in order to operate comes from households. Likewise, the products that households need and want come from businesses, and the factors of production that businesses need in order to operate come from households.

The circular flow model concentrates on the relationship between the two primary groups of actors in the economy: households and businesses. Both households and businesses take in money, and both of them spend money on goods and services. Businesses and households interact with one another in two kinds of markets: the product markets and the factor markets.

The idea that businesses make money by selling products to households is probably familiar. Whether the products are tangible goods, such as lamps and automobiles, or intangible services, such as medical care or accounting work, businesses make money by exchanging these products for money with members of households. The exchanges are regulated by the forces of supply and demand. At the top half of the circular flow model, products move in one direction, from businesses to households, and income moves in the other direction, from households to businesses. This top half of the model encompasses the product markets.

The money that flows into the hands of businesses is not absorbed or simply stored by businesses, however. In order to continue functioning, businesses must spend money on the factors of production: land (including physical land and any natural resources needed in their operations), labor, and capital (money, buildings, equipment, and other goods). Businesses use the money they obtain in the products market to buy these goods and services from households on the factor markets. In the case of labor, it may be obvious that this is so. One or more members of most households work in order to pay for the things they and their family members need and want. Workers sell their labor to businesses in much the same way that businesses sell products to households. Thus, the money that households contribute

to businesses in the form of purchases comes back to households in the form of payment for labor.

Less obviously, households ultimately sell the other factors of production to businesses as well. One of the main ways that households do this is by saving money in banks. When a worker takes a portion of his or her income and puts it in a savings account, the bank pays the worker interest. Meanwhile, the bank lends the worker's income out to businesses that need it in order to buy land or to make purchases of capital, such as factory equipment, trucks, or computers. The businesses must pay the bank interest on the money they borrow. In this situation the bank is no more than an intermediary: the worker is ultimately lending money to the business for a fee. In other words, the household is selling its money, at a price, to the business.

The basic two-sector (households and businesses), two-market (products and factors) circular flow model has become a classic illustration of economic activity. It is usually presented in the early pages of introductory economics textbooks. As a simple way of understanding the interaction of businesses and households, it is still considered an effective tool. Economists, however, regularly attempt to fill in missing details and make the circular flow model more precise. Accordingly, there are numerous variations on the basic diagram. One variation retains the two-sector focus but adds a third market, the financial market, to the product and factor markets. Another alternative diagram adds a third sector, government, and attempts to account for taxes in the circular flow. Yet another appends to both these diagrams the foreign sector, yielding a four-sector, three-market model. Still other economists attempt to build circular flow models that account for other factors, such as environmental damage and the monetary costs that it imposes on an economy.

Demand Analysis

Introduction & Meaning: Demand in common parlance means the desire for an object. But in economics demand is something more than this. According to Stonier and Hague, “Demand in economics means demand backed up by enough money to pay for the goods demanded”. This means that the demand becomes effective only if it is backed by the purchasing power in addition to this there must be willingness to buy a commodity. Thus demand in economics means the desire backed by the willingness to buy a commodity and the purchasing power to pay. In the words of “Benham” “The demand for anything at a given price is the amount of it which will be bought per unit of time at that Price”. (Thus demand is always at a price for a definite quantity at a specified time.) Thus demand has three essentials – price, quantity demanded and time. Without these, demand has no significance in economics. It deals with four aspects:

1. Consumption
2. Production
3. Exchange
4. Distribution

Factors Affecting Demand

There are factors on which the demand for a commodity depends. These factors are economic, social as well as political factors. The effect of all the factors on the amount demanded for the commodity is called Demand Function. These factors are as follows:

1. Price of the Commodity: The most important factor-affecting amount demanded is the price of the commodity. The amount of a commodity demanded at a particular price is more properly called price demand. The relation between price and demand is called the Law of Demand. It is not only the existing price but also the expected changes in price, which affect demand.

2. Income of the Consumer: The second most important factor influencing demand is consumer income. In fact, we can establish a relation between the consumer income and the demand at different levels of income, price and other things remaining the same. The demand for a normal commodity goes up when income rises and falls down when income falls. But in case of Giffen goods the relationship is the opposite.

3. Prices of related goods: The demand for a commodity is also affected by the changes in prices of the related goods also. Related goods can be of two types:

(i). Substitutes which can replace each other in use; for example, tea and coffee are substitutes. The change in price of a substitute has effect on a commodity's demand in the same direction in which price changes. The rise in price of coffee shall raise the demand for tea;

(ii). Complementary goods are those which are jointly demanded, such as pen and ink. In such cases complementary goods have opposite relationship between price of one commodity and the amount demanded for the other. If the price of pens goes up, their demand is less as a result of which the demand for ink is also less. The price and demand go in opposite direction. The effect of changes in price of a commodity on amounts demanded of related commodities is called Cross Demand.

4. Tastes of the Consumers: The amount demanded also depends on consumer's taste. Tastes include fashion, habit, customs, etc. A consumer's taste is also affected by advertisement. If the taste for a commodity goes up, its amount demanded is more even at the same price. This is called increase in demand. The opposite is called decrease in demand.

5. Wealth: The amount demanded of commodity is also affected by the amount of wealth as well as its distribution. The wealthier are the people; higher is the demand for normal commodities. If wealth is more equally distributed, the demand for necessities and comforts is more. On the other hand, if some people are rich, while the majorities are poor, the demand for luxuries is generally higher.

6. Population: Increase in population increases demand for necessities of life. The composition of population also affects demand. Composition of population means the proportion of young and old and children as well as the ratio of men to women. A change in composition of population has an effect on the nature of demand for different commodities.

7. Government Policy: Government policy affects the demands for commodities through taxation. Taxing a commodity increases its price and the demand goes down. Similarly, financial help from the government increases the demand for a commodity while lowering its price.

8. Expectations regarding the future: If consumers expect changes in price of commodity in future, they will change the demand at present even when the present

price remains the same. Similarly, if consumers expect their incomes to rise in the near future they may increase the demand for a commodity just now.

9. Climate and weather: The climate of an area and the weather prevailing there has a decisive effect on consumer's demand. In cold areas woolen cloth is demanded. During hot summer days, ice is very much in demand. On a rainy day, ice cream is not so much demanded.

10. State of business: The level of demand for different commodities also depends upon the business conditions in the country. If the country is passing through boom conditions, there will be a marked increase in demand. On the other hand, the level of demand goes down during depression.

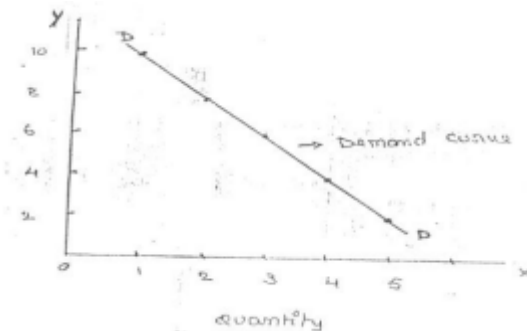
Law of Demand

Law of demand shows the relation between price and quantity demanded of a commodity in the market. In the words of Marshall, "the amount demand increases with a fall in price and diminishes with a rise in price". A rise in the price of a commodity is followed by a reduction in demand and a fall in price is followed by an increase in demand, if a condition of demand remains constant. The law of demand may be explained with the help of the following demand schedule.

Demand Schedule.

Price of Apple (In. Rs.)	Quantity Demanded
10	1
8	2
6	3
4	4
2	5

Law is demand is based When the price falls from Rs. 10 to 8 quantity demand increases from 1 to 2. In the same way as price falls, quantity demand increases on the basis of the demand schedule we can draw the demand curve.

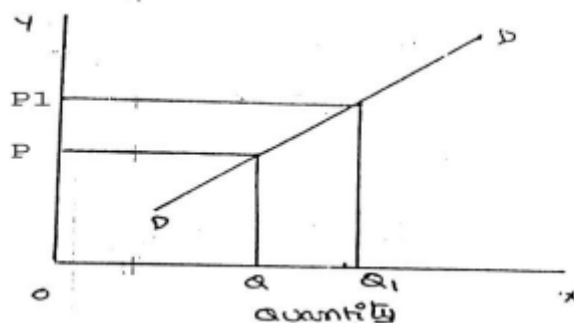


The demand curve DD shows the inverse relation between price and quantity demand of apple. It is downward sloping.

Assumptions:

1. This is no change in consumers taste and preferences.
2. Income should remain constant.
3. Prices of other goods should not change.
4. There should be no substitute for the commodity
5. The commodity should not confer at any distinction
6. The demand for the commodity should be continuous
7. People should not expect any change in the price of the commodity

Exceptional Demand Curve (Exceptions to Law of Demand): Sometimes the demand curve slopes upwards from left to right. In this case the demand curve has a positive slope.



When price increases from OP to Op1 quantity demanded also increases from to OQ1 and vice versa. The reasons for exceptional demand curve are as follows.

1. Giffen paradox: The Giffen good or inferior good is an exception to the law of demand. When the price of an inferior good falls, the poor will buy less and vice versa. For example, when the price of maize falls, the poor are willing to spend more on superior goods than on maize if the price of maize increases, he has to increase the quantity of money spent on it. Otherwise he will have to face starvation. Thus a fall in price is followed by reduction in quantity demanded and vice versa. “Giffen” first explained this and therefore it is called as Giffen’s paradox.

2. Veblen or Demonstration effect: „Veblen“ has explained the exceptional demand curve through his doctrine of conspicuous consumption. Rich people buy certain good because it gives social distinction or prestige for example diamonds are bought by the richer class for the prestige it possess. If the price of diamonds falls poor also will buy it hence they will not give prestige. Therefore, rich people may stop buying this commodity.

3. Ignorance: Sometimes, the quality of the commodity is judged by its price. Consumers think that the product is superior if the price is high. As such they buy more at a higher price.

4. Speculative effect: If the price of the commodity is increasing the consumers will buy more of it because of the fear that it increase still further, Thus, an increase in price may not be accomplished by a decrease in demand.

5. Fear of shortage: During the times of emergency of war People may expect shortage of a commodity. At that time, they may buy more at a higher price to keep stocks for the future.

6. Necessaries: In the case of necessities like rice, vegetables etc. people buy more even at a higher price.

Elasticity of Demand

Elasticity of demand explains the relationship between a change in price and consequent change in amount demanded. “Marshall” introduced the concept of elasticity of demand. Elasticity of demand shows the extent of change in quantity demanded to a change in price.

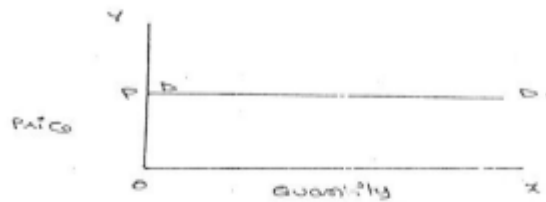
In the words of “Marshall”, “The elasticity of demand in a market is great or small according as the amount demanded increases much or little for a given fall in the price and diminishes much or little for a given rise in Price”

Elastic demand: A small change in price may lead to a great change in quantity demanded. In this case, demand is elastic.

In-elastic demand: If a big change in price is followed by a small change in demanded then the demand is “inelastic”.

Measurement of Elasticity of Demand

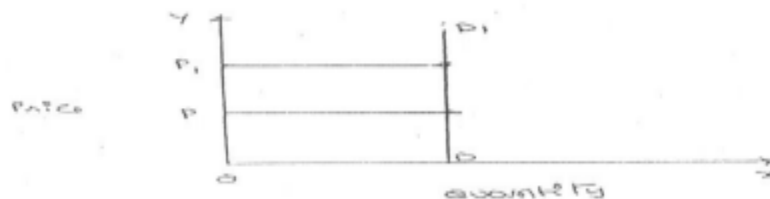
A. Perfectly elastic demand: When small change in price leads to an infinitely large change in quantity demanded, it is called perfectly or infinitely elastic demand. In this case $E = \infty$



The demand curve DD_1 is horizontal straight line. It shows that at “OP” price any amount is demanded and if price increases, the consumer will not purchase the commodity.

B. Perfectly Inelastic Demand

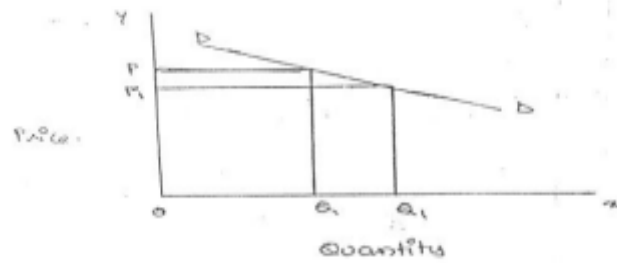
In this case, even a large change in price fails to bring about a change in quantity demanded.



When price increases from „OP” to „OP¹”, the quantity demanded remains the same. In other words the response of demand to a change in Price is nil. In this case „E”=0.

C. Relatively elastic demand:

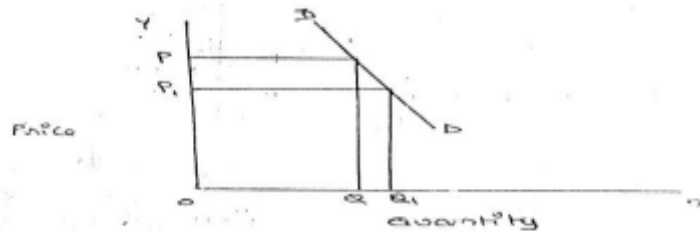
Demand changes more than proportionately to a change in price. I.e. a small change in price leads to a very big change in the quantity demanded. In this case $E > 1$. This demand curve will be flatter.



When price falls from „OP“ to „OP“, amount demanded increase from “OQ” to “OQ1” which is larger than the change in price.

D. Relatively in-elastic demand.

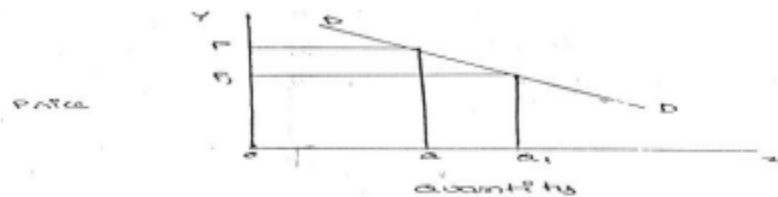
Quantity demanded changes less than proportional to a change in price. A large change in price leads to small change in amount demanded. Here $E < 1$. Demanded curve will be steeper.



When price falls from “OP” to „OP1 amount demanded increases from OQ to OQ1, which is smaller than the change in price

E. Unit elasticity of demand:

The change in demand is exactly equal to the change in price. When both are equal $E=1$ and elasticity is said to be unitary.



When price falls from „OP“ to „OP1“ quantity demanded increases from „OQ“ to „OQ1“. Thus a change in price has resulted in an equal change in quantity demanded so price elasticity of demand is equal to unity.

Types of Elasticity of Demand

1. Price Elasticity of Demand:

Marshall was the first economist to define price elasticity of demand. Price elasticity of demand measures changes in quantity demanded to a change in Price. It is the ratio of percentage change in quantity demanded to a percentage change in price.

$$\text{Price elasticity} = \frac{\text{Proportionate change in the quantity demand of commodity}}{\text{Proportionate change in the price of commodity}}$$

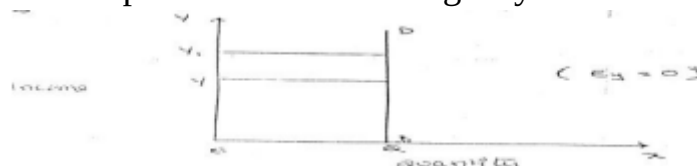
2. Income Elasticity of Demand:

Income elasticity of demand shows the change in quantity demanded as a result of a change in income. Income elasticity of demand may be slated in the form of a formula.

$$\text{Income Elasticity} = \frac{\text{Proportionate change in the quantity demand of commodity}}{\text{Proportionate change in the income of the people}}$$

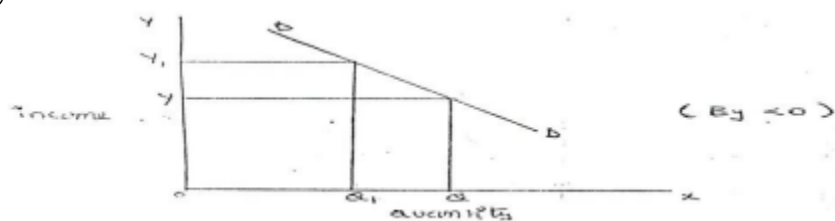
Income elasticity of demand can be classified in to five types.

A. Zero income elasticity: Quantity demanded remains the same, even though money income increases. Symbolically, it can be expressed as $E_y = 0$. It can be depicted in the following way:



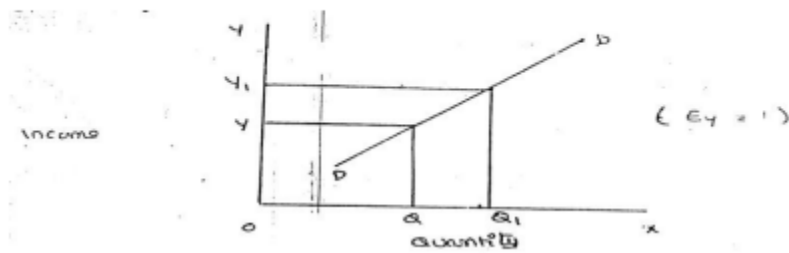
As income increases from OY to OY_1 , quantity demanded never changes.

B. Negative Income elasticity: When income increases, quantity demanded falls. In this case, income elasticity of demand is negative. i.e., $E_y < 0$.



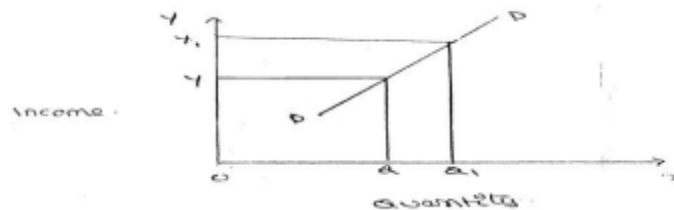
When income increases from OY to OY_1 , demand falls from OQ to OQ_1 .

C. Unit income elasticity: When an increase in income brings about a proportionate increase in quantity demanded, and then income elasticity of demand is equal to one. $E_y = 1$



It shows high-income elasticity of demand. When income increases from OY to OY_1 , Quantity demanded increases from OQ to OQ_1 .

D. Income elasticity less than unity: When income increases quantity demanded also increases but less than proportionately. In this case $E < 1$.



An increase in income from OY to OY_1 , brings what an increase in quantity demanded from OQ to OQ_1 , But the increase in quantity demanded is smaller than the increase in income. Hence, income elasticity of demand is less than one.

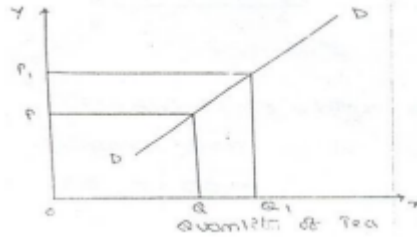
3. Cross Elasticity of Demand:

A change in the price of one commodity leads to a change in the quantity demanded of another commodity. This is called a cross elasticity of demand. The formula for cross elasticity of demand is:

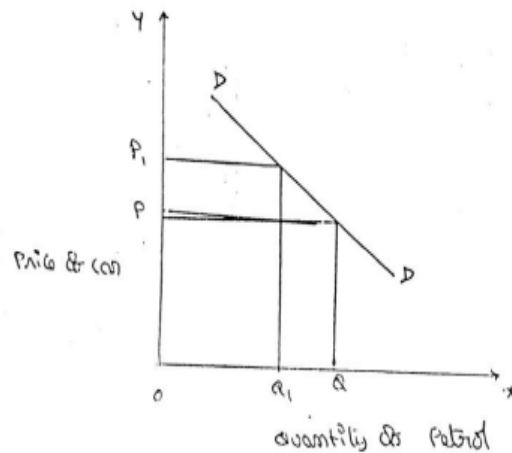
Proportionate change in the quantity demand of commodity "X"

Cross elasticity = $\frac{\text{Proportionate change in the quantity demand of commodity "X"}}{\text{Proportionate change in the price of commodity "Y"}}$

A. In case of substitutes, cross elasticity of demand is positive. E.g.: Coffee and Tea When the price of coffee increases, Quantity demanded of tea increases. Both are substitutes.

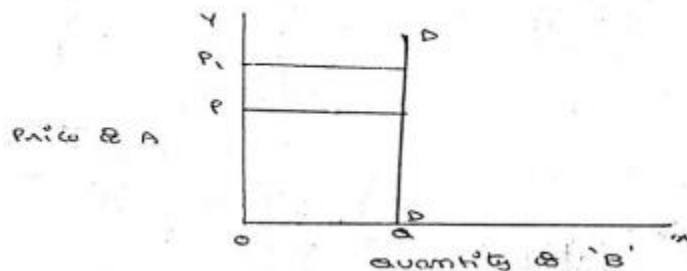


B. In case of compliments, cross elasticity is negative. If increase in the price of one commodity leads to a decrease in the quantity demanded of another and vice versa. When price of car goes up from OP to OP₁, the quantity demanded of petrol decreases from OQ to OQ₁!. The cross-demanded curve has negative slope.



$$E_{cd} = \frac{\% \Delta Q_1}{\% \Delta P_1} \text{ (Negative)}$$

C. In case of unrelated commodities, cross elasticity of demanded is zero. A change in the price of one commodity will not affect the quantity demanded of another.



Quantity demanded of commodity "b" remains unchanged due to a change in the price of „A“, as both are unrelated goods.

4. Advertising Elasticity of Demand: Advertising elasticity of demand shows the change in quantity demanded as a result of a change in cost of Advertisement. Advertising elasticity of demand may be slated in the form of a formula

$$\text{Advertising Elasticity} = \frac{\text{Proportionate change in the quantity demand of commodity}}{\text{Proportionate change in the advertisement cost}}$$

Market equilibrium price

Equilibrium is the state in which market supply and demand balance each other, and as a result prices become stable. Generally, an over-supply of goods or services causes prices to go down, which results in higher demand—while an under-supply or shortage causes prices to go up resulting in less demand.

The balancing effect of supply and demand results in a state of equilibrium.

KEY TAKEAWAYS

- A market is said to have reached equilibrium price when the supply of goods matches demand.
- A market in equilibrium demonstrates three characteristics: the behavior of agents is consistent, there are no incentives for agents to change behavior, and a dynamic process governs equilibrium outcomes.
- There are several types of equilibrium used in economics.
- Disequilibrium is the opposite of equilibrium and it is characterized by changes in conditions that affect market equilibrium.
- In reality, markets are never in perfect equilibrium, although prices do tend toward it.

The equilibrium price is where the supply of goods matches demand. When a major index experiences a period of consolidation or sideways momentum, it can be said that the forces of supply and demand are relatively equal and the market is in a state of equilibrium.

Economists find that prices tend to fluctuate around the equilibrium levels. If the price rises too high, market forces will incentivize sellers to come in and produce more. If the price is too low, additional buyers will bid up the price. These activities keep the equilibrium level in relative balance over time.

Economists like Adam Smith believed that a free market would tend toward equilibrium. For example, a dearth of any one good would create a higher price generally, which would reduce demand, leading to an increase in supply provided the right incentive. The same would occur in reverse order provided there was excess in any one market.

Modern economists point out that cartels or monopolistic companies can artificially hold prices higher and keep them there in order to reap higher profits. The diamond industry is a classic example of a market where demand is high, but supply is made artificially scarce by companies selling fewer diamonds in order to keep prices high.

Types of Equilibrium:

1. **Economic Equilibrium:** Economic equilibrium refers broadly to any state in the economy where forces are balanced. This can be related to prices in a market where supply is equal to demand, but can also represent the level of employment, interest rates, and so on.
2. **Competitive Equilibrium:** The process by which equilibrium prices are reached is through a process of competition. Among sellers to be the low-cost producer to grab the largest market share, and also among buyers to snatch up the best deals.
3. **General Equilibrium:** General equilibrium considers the aggregation of forces occurring at the macro-economic level, and not the micro forces of individual markets. It is a cornerstone of Walrasian economics.
4. **Underemployment Equilibrium:** Economists have found that there is a level of persistent unemployment that is observed when there is general equilibrium in an economy. This is known as underemployment equilibrium, and is predicted by Keynesian economic theory.
5. **Lindahl Equilibrium:** Lindahl equilibrium is a special case where, in theory, the optimal amount of public goods is produced and the cost of public goods is fairly shared among everyone. It describes an ideal state rarely, if ever, achieved in reality, but is used to help craft tax policy and is an important concept in welfare economics.

6. **Intertemporal Equilibrium:** Because prices may swing above or below the equilibrium level due to proximate changes in supply or demand at a given moment, it is best to look at this effect over time, known as intertemporal equilibrium. The concept is also used in understanding how firms and households budget and smooth spending over longer time horizons.
7. **Nash Equilibrium:** In game theory, Nash equilibrium is a state of play whereby the optimal strategy involves considering the optimal strategy of the other player or opponent.

Example of Equilibrium

A store manufactures 1,000 spinning tops and retails them at \$10 per piece. But no one is willing to buy them at that price. To pump up demand, the store reduces its price to \$8. There are 250 buyers at that price point. In response, the store further slashes the retail cost to \$5 and garners five hundred buyers in total. Upon further reduction of the price to \$2, one thousand buyers of the spinning top materialize. At this price point, supply equals demand. Hence \$2 is the equilibrium price for the spinning tops.